

WhatsApp Receipt Automation System – Setup Guide

Overview

This system automates the entire process of:

1. Receiving WhatsApp messages with receipt images
2. Extracting text and key transaction data using OCR
3. Detecting the supplier and bank automatically
4. Uploading the image to Google Drive (in the right folder)
5. Writing the data into Google Sheets (with dynamic sheet creation)

Once deployed, the system runs 24/7 — no need to keep your PC open.

1. System Components

Your project contains the following core services:

Service	Description
FastAPI	Receives messages from WhatsApp webhook
Celery Worker	Handles background tasks (OCR, upload, sheet writing)
Redis	Message broker for Celery
WhatsApp Listener (Node.js)	Listens for incoming WhatsApp messages and forwards them to FastAPI
Google Drive Integration	Uploads receipts into correct supplier folders
Google Sheets Integration	Logs transaction data into appropriate sheet

2. Required Accounts

You'll need:

- A **Meta Developer Account** (for WhatsApp API)
- A **Google Cloud Project** with OAuth 2.0 credentials
- A **Google Drive folder** (will contain subfolders per supplier)

- A **Google Sheet** (will auto-expand with new sheets if needed)
- A **Render Account** (for deployment)

3. Environment Variables

Before deploying, set these variables in **Render** → **Environment**:

Variable	Description
GOOGLE_CLIENT_ID	From Google Cloud OAuth credentials
GOOGLE_CLIENT_SECRET	From Google Cloud OAuth credentials
GOOGLE_TOKEN	Short-lived token obtained during auth
GOOGLE_REFRESH_TOKEN	Refresh token for long-term access
DRIVE_FOLDER_ID	ID of the main parent folder in Google Drive
REDIS_URL	Redis connection string
META_VERIFY_TOKEN	Your custom Meta verification token
META_ACCESS_TOKEN	WhatsApp Cloud API access token

Tip: use the included `.env_template.txt` to organize your values.

4. Folder Structure in Google Drive

The bot automatically creates and uploads receipts into the following folders inside your **main parent folder**:

1. **General Suppliers**
 - Transgestiona SA ◦ Prestigio Pagos SA ◦ Plataforma de Pago SA
 - Aurinegros SA
 - Cobro Sur SA
2. **Cobro Express SA**
3. **CLAN SRL**

4. RAZ Y CIA

5. **Others** – all receipts not matching any supplier above

If the folder already exists, it won't be recreated.

5. Google Sheets Automation

Each supplier has its own sheet in Google Sheets.

When a new sheet is created, headers are automatically added.

If one sheet fills up, a new sheet is created automatically — maintaining structure and alignment.

Each row includes:

Column	Description
Date	Extracted from receipt
Amount	Detected transaction amount
Supplier	Detected supplier name
Destination Bank	Detected bank (standardized)
Image Link	Public link to receipt in Drive
Transaction ID	Unique transaction reference
Timestamp	System time of entry

6. WhatsApp Integration

Once deployed, connect your WhatsApp Business Number:

1. Go to your **Meta Developer Dashboard**
2. Select your app → WhatsApp → Configuration
3. Add webhook:
Callback URL: <https://your-app-name.onrender.com/webhook>
Verify Token: same as your META_VERIFY_TOKEN
4. Subscribe to the **messages** field under Webhooks.

Send a test message with a receipt image — the bot will process it and log the result automatically.

7. Deployment on Render You will deploy **three services**:

Service	Type	Start Command
FastAPI	Web Service	<code>python -m uvicorn main:app --host 0.0.0.0 --port 8000</code>
Celery Worker	Background Worker	<code>celery -A tasks worker --loglevel=info</code>
WhatsApp Listener	Background Worker (Node.js)	<code>node listener.js</code>

Once deployed:

- The system stays live even if your PC is off.
- Logs are available under each service on Render.

8. File Cleanup & Maintenance

- Temporary images are deleted after upload to keep the server clean.
- Uploaded files remain safely stored in your Google Drive.
- Sheet data is never deleted unless done manually.